SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	CSA0615	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO5– Naturalization (BL5,BL6)	**Assessment:**	Assessment Tool 1 – Constraint Based Problem Solving
Weightage:	30% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO5 Statement:	*Analyze computational problems using backtracking techniques and complexity classes such as P, NP, NP-Hard, and NP-Complete for combinatorial problem solving.*		
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Section / Batch:	B.TECH AI&DS	**Date:**	31-08-2026

Instructions to Students

• • Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
• Apply backtracking techniques systematically for combinatorial problem solving.
• Clearly classify problems under P, NP, NP-Hard, and NP-Complete categories wherever required. • Provide proper justification, state-space representation, and complexity analysis for all solutions. • Neat presentation and logical explanation are mandatory.

Question Type	Focus Area	Questions	Marks
Constraint Based Problem Solving	Analyze combinatorial problems using backtracking and complexity classes such as P, NP, NP-Hard, and NP Complete.	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:	**100 Marks**	

SECTION A – CONSTARINT BASED PROBLEM SOLVING

SIMATS ENGINEERING

SIMATS Online Programming Lab

DAA PROGRAMMING

CO5_AT1_Constraint-Based Problem Solving

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Language: Python | C | C++ | Java

QUESTIONS

Q1 — Backtracking and Tractability — 10 marks

Q2 — Backtracking and Tractability — 10 marks

1/2 answered

Q1 Backtracking and Tractability — 10 marks

Knight's Tour Problem (Backtracking / Constraint Satisfaction Problem)

A knight must visit every square of a chessboard exactly once. Clearly define constraints related to valid knight moves and non-repetition. Analyze how these constraints reduce possible moves at each step. Develop a backtracking-based solution to explore paths. Apply pruning to eliminate dead-end paths. Justify how your solution ensures completeness and discuss complexity.

Your Code

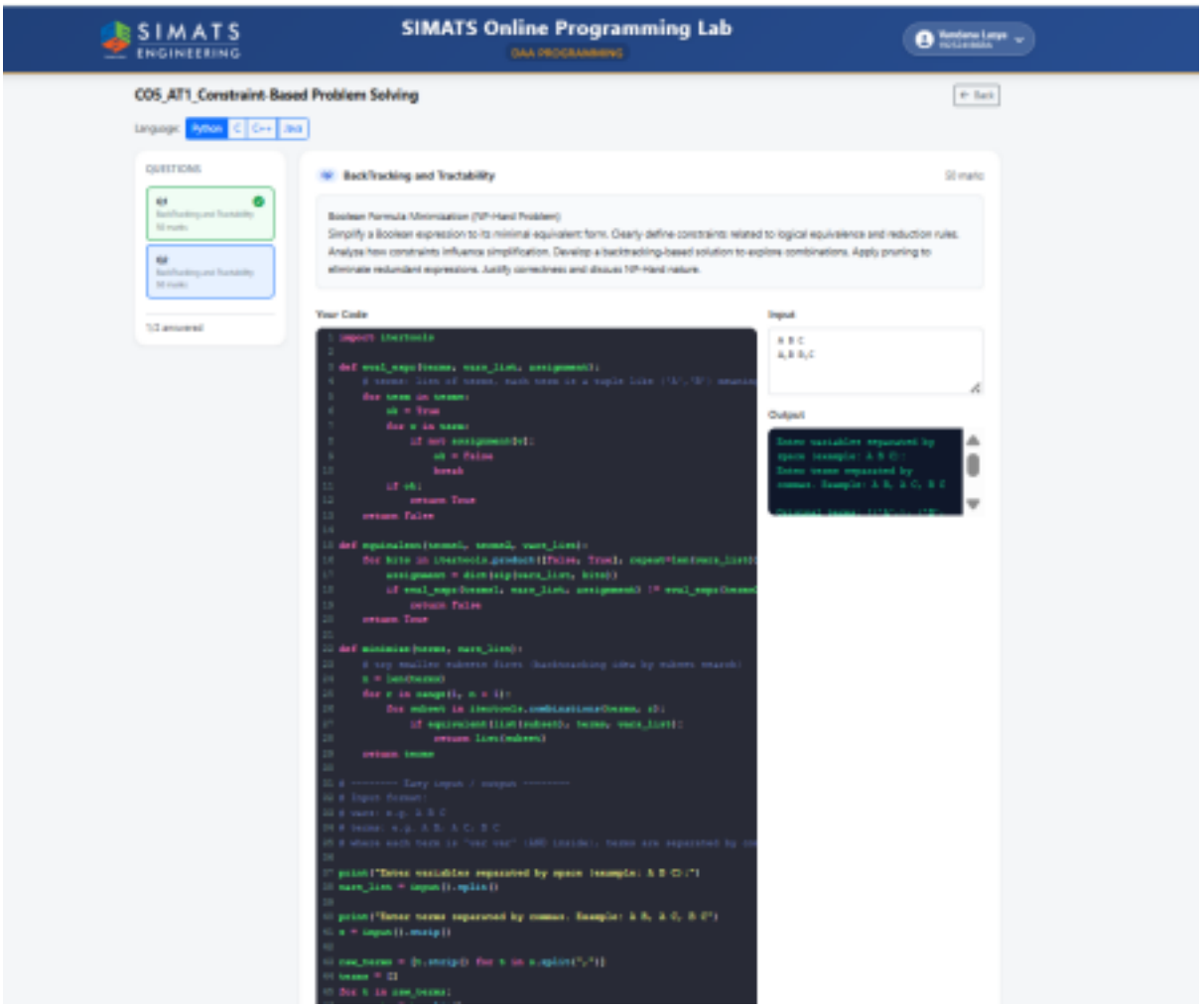
```python
solve_knights_tour()
# 1. Get user input
try:
    n = int(input("Enter board size (e.g., 5 or 8): "))
except ValueError:
    print("Please enter a valid number.")
    return

# 2. Initialize the board with -1 (representing unvisited)
board = [[-1 for _ in range(n)] for _ in range(n)]

# 3. Define the 8 possible moves a knight can make
move_x = [2, 1, -1, -2, -2, -1, 1, 2]
move_y = [1, 2, 2, 1, -1, -2, -2, -1]

# Starting position
board[0][0] = 0

def is_safe(x, y, board):
    # Check if move is inside board and square is unvisited
    return 0 <= x < n and 0 <= y < n and board[x][y] == -1

def backtrack(curr_x, curr_y, move_count):
    # Base Case: If all squares are visited, we are done!
    if move_count == n * n:
        return True

    # Try all 8 possible moves
    for i in range(8):
        next_x = curr_x + move_x[i]
        next_y = curr_y + move_y[i]

        if is_safe(next_x, next_y, board):
            board[next_x][next_y] = move_count

            # Recursive step: try to find the next move
            if backtrack(next_x, next_y, move_count + 1):
                return True

            # BACKTRACKING: If move doesn't work, undo it (reset to -1)
            board[next_x][next_y] = -1

    return False

# 4. Run the solver
if backtrack(0, 0, 1):
```

Input

```
5
```

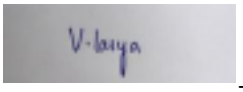
Output

```
Enter board size (e.g., 5 or
8):
Success! Here is the Knight's
Tour path:
 0  9 14  5 20
```



Code of Conduct

I _____V.LASYA(192524186)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:  _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understa ndin g	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understandi ng; misses key constraints	Fails to identify constraints correctly
Constrain t Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritizatio n	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Develop ment	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints

Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Marks Evaluation:

Question No.	Problem Understanding (25%)	Constraint Analysis (25%)	Solution Development (25%)	Application of Concepts (15%)	Justification (10%)	Total Marks (Each – 50)
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____